



Effects of common prosperity on China's education expenditure—Empirical analysis based on CFPS quasi-micro panel data

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ABSTRACT

What effects do decreased income gaps have on promoting a high-quality economy under gradual common prosperity in China? The work initially answered the question by focusing on education expenditure. It is a form of consumption that has great significance to the future development of residents and overall social and technological innovation. The impact mechanism and pathway of income gaps on education expenditure were systematically analyzed from a theoretical level in the work. The nonlinear relationship between the two was empirically verified based on CFPS quasi-micro panel data. The law of diminishing marginal propensity to consume in mainstream economics could be challenged when high-income families focused on the investment effects of education and social status-seeking of consumption. The same could be true when low-income countries focused on the investment effects of education and the catch-up effects of consumption. Consequently, changed income gaps had an inverted U-shaped impact on social education expenditure. The consumption propensity of high-income families was higher than that of low-income ones if income gaps were smaller than the fixed consumption distance. Widened income gaps increased social education expenditure at this time, and vice versa. The empirical analysis further found that income gaps in most districts (counties) in China exceeded the critical value during the sample period based on the verification of an inverted U-shaped relationship. Therefore, narrowed income gaps by common prosperity had no significant inhibitory effect on education expenditure and human capital accumulation. The critical value would maximize overall education expenditure to promote a high-quality economy in China.

1. Introduction

Common prosperity is one of the key elements of Deng Xiaoping's theory of building socialism with Chinese characteristics, specifically referring to the fact that all people, through hard work and mutual assistance, will eventually achieve a standard of living that is well-fed and well-coated, i.e., universal prosperity on the basis of the elimination of polarisation and poverty, but such universal prosperity is not, nor can it be, achieved through the equal distribution of existing wealth. After more than 40 years of reform and development, China has made great strides in liberating and developing the productive forces and promoting common prosperity, with

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the general public benefiting more and more from reform and development, the number of people living in poverty in society decreasing significantly, and living standards improving to varying degrees. At the same time, the trend of widening income disparities between regions, between urban and rural areas, and between different groups has not yet been effectively curbed, and the irrational pattern of distribution of national income has seriously undermined the people's motivation, making the promotion of common prosperity on the basis of economic development a major practical issue that must be seriously addressed in the development of socialism with Chinese characteristics.

China's income-distribution inequality has continued to increase with sustained and rapid economic growth over the past 40 years of reform and opening up. Income inequality still exists widely between urban and rural areas, regions, and industries despite the government's efforts to unify fairness and efficiency. Income gaps have increased, showing a trend of further widening (Jian, 2018). The current situation, development trend and causes of China's income disparity are summarized as follows: China's current income disparity is high, and the overall situation shows a widening trend. Factors such as the system of fiscal revenues and expenditures, the degree of monopoly and rent-seeking behaviours in industries, the tax structure, the dual economic structure of urban and rural areas, the calibre and scope of statistics, the level of financial deepening, and the marketization of returns on factors of production, etc. are the main reasons for the high and widening income disparity in China. 3. The main reason for the high and widening income gap in China. China's Gini coefficients were 0.491 and 0.462 in 2003 and 2021, respectively, which were far higher than the international warning level of 0.4. The top 1% of high-wealth families owned more than one-third of China's property, while the bottom 25% of low-wealth ones owned only about 1% of that considering property distribution from 1995 to 2021.

The proportion of wage income continues to decline considering income distribution. The proportion of property income increased from 2.5 to 8.8%, and the proportion of wage income decreased from 72.1 to 55.9% from 2009 to 2021. Property income grew by 10.2% in 2021 and was higher than 9.6% of wage income. It further declined (data from *China Yearbook of Household Survey*, *China Family Panel Studies* (CFPS), and *Annual Report on China Household Income Distribution 2021*). China's income distribution inequality is significantly increasing, and overall income distribution is relatively unequal. Common prosperity to narrow income gaps in China has become a general trend. The report to the 20th National Congress of the CPC stated that one of the overall goals of China is to achieve more obvious substantive progress in common prosperity by 2035.

Advanced common prosperity and continuously narrowed income gaps may have a series of impacts on high-quality development. One of the most important impact pathways is residents' education expenditure. Education is an important means to improve human capital accumulation and plays a vital role in a country's technological progress and high-quality development (Zhang et al., 2022). Increased education expenditure improves personal well-being and family social status. It increases human capital accumulation and promotes technological innovation for the realization of high-quality development (Zhang et al., 2022). Fully understanding the relationship between income inequality and education expenditure is crucial. Income inequality and education expenditure show a continuous growth trend before 2018. Widened income gaps are generally believed to have a significant promoting effect on increasing residents' education expenditure (Wu et al., 2018; Yan & Hang, 2017; Zhang, 2019). Yan and Hang (2017) analyzed the education expenditure of Chinese families in compulsory education concerning status based on the relative income hypothesis using CFPS. The expansion of income inequality strengthens families' status-focused motivations and further pushes up the overall level of family education expenditure. Wu et al. (2018) proposed that low-income families have strong incentives to improve their income class by increasing education investment under income inequality. Therefore, income gaps have a significant promoting effect on family educational consumption. Zhang (2019) integrated the life cycle hypothesis and social-status seeking assuming non-independence preferences in his theoretical analysis. Current family education expenditure is an investment related to the family's future social status. Widening income gaps have strengthened the demonstration effect and triggered competition in education expenditure, which ultimately increases the overall level of household education expenditure.

The studies mainly focus on the consumption behaviors of low-income families and do not pay attention to the consumption-behavior choices of high-income ones facing changes in income gaps, social status, and class mobility. High-income families often take the initiative in consumer behavior choices in reality because they have economic and information advantages. It is an important factor affecting overall educational consumption. The Gini coefficient of per capita disposable income and per capita education expenditure showed a growth trend before 2018 according to the *China Yearbook of Household Survey*. The existing theoretical hypothesis seems to be verified that widening income gaps can increase education expenditures. Relevant statistical data in recent years presents contradictory findings. The relationship between income-gap changes and education expenditures has been reversed for three consecutive years after 2018 according to the *China Yearbook of Household Survey*. The Gini coefficient of per capita disposable income dropped from 0.468 to 0.465 in 2019, while per capita education expenditure increased from 1398.3 to 1664.5 yuan. The Gini coefficient of per capita disposable income increased from 0.465 to 0.468 in 2020, while per capita education expenditure dropped from 1664.5 to 1463.2 yuan. The Gini coefficient of per capita disposable income dropped from 0.468 to 0.466 in 2021, while per capita education expenditure increased from 1463.2 to 1953.2 yuan.

Increased education expenditure runs counter to widened income gaps in the context of fluctuated income gaps, especially concerning common prosperity. The anomalous phenomenon, which contradicts the existing theoretical hypothesis, has prompted further contemplation on the correlation between altered income disparities and education expenditure. Widening income gaps is assumed to increase education expenditure according to the existing theoretical hypothesis. Narrowed income gaps will have a significant inhibitory effect on education expenditure and human capital accumulation with advanced common prosperity. It is not conducive to the high-quality development of the overall economy. Common prosperity to narrow income gaps may harm high-quality economic development. Therefore, the necessity and urgency of common prosperity will be greatly reduced. However, official statistics show that the relationship between income gaps and education expenditure seems to be non-linear.

There may be an inflection point in the promoting effect of widening income gaps on increasing education expenditure. How does

the nonlinear relationship emerge? Why is there an inflection point? The specific mechanism and pathway of income gaps affecting education expenditure need interpreting theoretically. Education expenditure is a typical micro-family consumption behavior. Empirical verification of its relationship with varying income inequality should use micro-level data, especially recently popular micro-panel data. Currently, authoritative micro-family survey data are obtained through random sampling, which results in significant differences or even no correlation in individual families surveyed each year. It is difficult to obtain micro-level panel data for scientific empirical analysis.

CFPS microdata was used in the work to synthesize quasi-micro panel data of representative (average) households at the district (county) level to test the theoretical hypothesis. An empirical verification of the relationship between income disparities and education expenditure was performed more rigorously to enhance the scientific rigor of our findings. A mathematical model was developed to analyze the impact of varying income disparities on education expenditures, incorporating fundamental theories such as the relative income hypothesis, positional goods, and status-seeking. These theoretical hypotheses were empirically validated using quasi-microscopic panel data from CFPS spanning 2012 to 2018. The law of diminishing marginal propensity to consume in mainstream economics was challenged when high-income families focused on the investment effects of education and social-status seeking of consumption. The same was true when low-income countries focused on the investment effects of education and the catch-up effects of consumption.

Consequently, changed income gaps have an inverted U-shaped impact on social education expenditure. The consumption propensity of high-income families is higher than that of low-income ones if income gaps are smaller than the fixed consumption distance. Widened income gaps increase social education expenditure at this time, and vice versa. The empirical analysis further shows that income gaps in most districts (counties) in China exceed the critical value during the sample period based on its verification of the inverted U-shaped relationship. Excessive income gaps exist in current China. Narrowed income gaps by common prosperity have no significant inhibitory effect on education expenditure and human capital accumulation. The critical value maximizes overall education expenditure to promote the high-quality economy of China.

The structure of the remaining is as follows. Section 2 shows the theoretical analysis and hypothesis. Section 3 presents data sources, variable measurement, and statistical analysis. Section 4 illustrates the econometric model and empirical results analysis. Conclusions and policy implications are listed in Section 5.

2. Theoretical analysis and hypothesis formulation

The section constructs a mathematical model of varying income gaps affecting education expenditures based on basic theories, e.g., the relative income hypothesis, positional goods, and status-seeking. The specific mechanism and pathway through which varying income gaps affect education expenditure are discovered, with corresponding theoretical hypotheses proposed.

2.1. Basic assumptions

(1) There are two types of families assumed in society—low-income and high-income families. w_1 and w_2 are the average income of low-income and high-income families, respectively. $1-\delta$ and δ are the proportions of the number of low-income and high-income families, respectively (δ is a constant; $0 < \delta < 1$). ine is the degree of income inequality as well as the ratio of the average income of high-income families to that of low-income ones.

$$ine = \frac{w_2}{w_1} \quad (1)$$

Exogenous variable Dy is introduced to study the impact of varying income gaps on education expenditure. Varying Dy will change ine (Eq. (2))

$$ine = \frac{w_2 - \left(\frac{1}{\delta}\right)Dy}{w_1 + \left(\frac{1}{1-\delta}\right)Dy} \quad (2)$$

(2) Education expenditure is a consumption and investment behavior, related to future income level and social status. Individuals often regard education as an investment to maximize long-term utility and maintain existing social status and wealth. The investment lasts throughout the entire life cycle of an individual, and we call it the investment effects of education expenditure. Notably, the influence degree of investment effects of education on individuals is not static and affected by wealth level, the return rate of other investment behaviors, etc. Investment effects of education expenditure will weaken when the wealth level is high or the return rate of educational investment is relatively low.

Household utility is assumed as a function of general commodity consumption and educational consumption, and its Cobb-Douglas production function $u_i = c_i^\alpha e_i^{1-\alpha}$ (Becker and Lindsay (1994)). c and e are the family's general commodity consumption and education expenditure, respectively ($0 < \alpha < 1$); $i = 1$ and 2 are low-income and high-income families, respectively. Educational investment is widely considered to be the most effective way to increase future consumption levels for enhancing household utility although there are many alternative ways to maximize household utility (Wang et al., 2021).

General commodity consumption satisfies current needs, while education expenditure can increase future income to improve future consumption levels and satisfaction. Therefore, families need to make trade-offs between general commodity consumption and

education consumption to maximize family utility. Households are assumed to be unable to obtain funds for education expenditure from the capital market at a lower cost due to its imperfection considering China's current reality. They can only make trade-offs between ordinary commodity consumption and education consumption based on actual income. Consequently, the income status of a family determines its education expenditure to a large extent (Becker & Tomers, 1979).

The total consumption of a family is the sum of c and e , which is equal to the product of the family's consumption propensity b_i ($i = 1$ and 2) and their respective family incomes. Budget constraints for the incomes of the two types of families are obtained as $c_1 + e_1 = b_1[w_1 + (\frac{1}{1-\delta})Dy]$ and $c_2 + e_2 = b_2[w_2 - (\frac{1}{\delta})Dy]$, respectively. Therefore, the high-income class hopes to improve family utility and demonstrate their social status by consuming more and higher-quality goods and education than the low-income ones. This leads to the last and most important hypothesis of this section.

(3) Consumer economics believes that people's consumption preferences are non-independent. There are two important concepts in the relative income hypothesis proposed by Duesenberry (1949)—the demonstration effect and the ratchet effect. The demonstration effect refers to that consumers are affected by the consumption levels of people around them. Consumers are affected by consuming environments, and their consumption preferences are not independent. The ratchet effect refers to characteristics of consumer behaviors that are “easy to ascend and hard to descend”. Consumer behavior is influenced by past consumption. Decreased income levels do not result in corresponding decreased consumption levels, and vice versa.

The demonstration effect and the ratchet effect have different impacts on high and low-income classes. The demonstration effect makes the high-income class hope to achieve it by consuming more goods and education than the low-income one. High-end goods and high-quality education are generally recognized positional goods. Consumption of them is considered to be a typical manifestation of social status. Therefore, the work summarized the impact of the demonstration effect on the high-income class as the status-seeking of consumer spending.

The high-income class in society hopes to determine and demonstrate its social status by consuming more goods and education than the low-income one regardless of levels of income disparity. Status seeking is thus fulfilled. The enduring impact of this phenomenon primarily targets individuals in the high-income class. The ratchet effect motivates the high-income class to maintain a stable long-term gap with the low-income one concerning commodity consumption and education expenditure. They are determined by the dual influence of investment effects and status-seeking.

The demonstration effect causes the low-income class to be influenced by the high-income one and imitate to catch up with them. They hope to catch up with the high-income class in terms of social status, wealth level, and life quality. Therefore, the work summarized the impact of the demonstration effect on the low-income class as the catch-up effect of consumer spending. The ratchet effect has caused the low-income class to make increasing commodity consumption and education consumption as the main ways to catch up with the high-income one and individual investment chronically due to the single investment channel, information lag, etc. Consequently, the level of commodity consumption and education consumption of the low-income class is “easy to ascend but difficult to descend” and even rising.

High-income families hope to accumulate human capital through more education consumption of higher quality than low-income ones and maintain their existing social status and wealth (investment effects of pursuing education (Wu et al., 2020)). They also aspire to attain status-seeking behavior (Qian, 2004; Yu, 2014). Most families in China currently view education as a form of investment and hope to demonstrate their social class and status by increasing consumer spending on goods and education. Economic status is an important aspect of social status from a consumption perspective. Households with high commodity consumption and education expenditure own high incomes; thus, they have a higher social status. Education is an objective standard for social hierarchy from the perspective of education expenditure or investment (Fallon, 1999; Jin Ye & Wu, 2011). Members of families with a higher average education level have higher levels of knowledge, skills, and moral qualities, and they are respected by the public. Therefore, the whole family has a high social status. Education is an important way of human capital accumulation and the reason for differences in individual income (Xue et al., 2008). Families with a higher average education level allow their members to receive more education of quality to maintain their existing economic and social status. Veblen's conspicuous consumption hypothesis mentioned that there are differences in consumption decisions between high-income and low-income groups. Consumption decisions of the high-income class are for a leisurely life and showing off their financial resources and social status to others. Conspicuous consumption is more important than a leisurely life. Hirsch proposed the concepts of positional goods and material goods in *Social Limits to Growth*.

Differences between positional goods and material goods are as follows. Positional goods are scarce, referring to material scarcity and social scarcity caused by social limitations. The utility of material goods is only related to the quantity of one's consumption, while that of positional goods depends on comparison with others or relative positions. There is a relative price or quality gap if measured by consumer price or consumer quality. Collins (2018), an American sociologist, provides a historical sociological analysis of the 'diploma society' in the United States, arguing that as long as some people have a higher level of education than others, they are likely to see education as a way of gaining social status, and consequently as a credentialing tool for social status. Yan and Hang (2017) believed that education has obvious attributes of positional goods. High-income families hope to achieve social hierarchy by consuming positional goods. Therefore, families with different income levels are extremely concerned about their differences concerning general commodity consumption and education consumption, which have become the key to social stratification.

There is fixed distance ω (ratio of general consumption between high-income and low-income families; $\omega > 1$) in terms of consumption quantities or quality based on the analysis and the consumption distance and fixed difference theory of Veblen et al. as well as the ratchet effect of consumption. The existence of ω can be attributed to the following factors. First, high-income families hope to accumulate human capital and maintain their existing social status and wealth through more education consumption of higher quality than low-income ones to pursue the investment effects of education. Second, they hope to achieve status-seeking through that.

There has been a long-term gap between high-income and low-income families concerning levels of commodity consumption and education consumption under the ratchet effect. This gap is relatively stable. Generally, the fixed distance is closely related to the popular concepts, traditional customs, and consumption levels of an era and society. Once formed, it persists for a long period and can therefore be considered exogenously given. In summary, high-income families hoped to pursue the investment effects of education and the status-seeking of consumption by maintaining a certain fixed consumption distance from low-income families.

The section systematically analyzes the total impact of varying income gaps on social education expenditure under the three assumptions.

2.2. Influences of varying income gaps on social education expenditures

The consumption decision of low-income families is defined by

$$\begin{aligned} \text{Max} \quad & u_1 = c_1^\alpha e_1^{1-\alpha} \\ \text{s.t.} \quad & c_1 + e_1 = b_1 \left[w_1 + \left(\frac{1}{1-\delta} \right) Dy \right] \end{aligned} \quad (3)$$

where c_1 is the commodity consumption of low-income families; e_1 is the family education consumption; b_1 is the consumption tendency of low-income families.

Therefore, a Lagrange function is established.

$$L = c_1^\alpha e_1^{1-\alpha} + \lambda_1 \left\{ b_1 \left[w_1 + \left(\frac{1}{1-\delta} \right) Dy \right] - c_1 - e_1 \right\} \quad (4)$$

The following first-order conditions are obtained (Eq. (5)).

$$\frac{\partial L}{\partial c_1} = \alpha c_1^{\alpha-1} e_1^{1-\alpha} - \lambda_1 = 0 \quad (5)$$

$$\frac{\partial L}{\partial e_1} = (1-\alpha) c_1^\alpha e_1^{-\alpha} - \lambda_1 = 0 \quad (6)$$

$$\frac{\partial L}{\partial \lambda_1} = b_1 \left[w_1 + \left(\frac{1}{1-\delta} \right) Dy \right] - c_1 - e_1 = 0 \quad (7)$$

Eqs. (5)–(7) are used to obtain

$$c_1 / e_1 = \alpha / (1-\alpha) \quad (8)$$

$$c_1 = \alpha b_1 \left[w_1 + \left(\frac{1}{1-\delta} \right) Dy \right] \quad (9)$$

$$e_1 = (1-\alpha) b_1 \left[w_1 + \left(\frac{1}{1-\delta} \right) Dy \right] \quad (10)$$

Consequently, the education consumption of low-income families is $e_1 = (1-\alpha) b_1 \left[w_1 + \left(\frac{1}{1-\delta} \right) Dy \right]$ to maximize family utility.

Consumption decisions of high-income families are defined by

$$\begin{aligned} \text{Max} \quad & u_2 = c_2^\alpha e_2^{1-\alpha} \\ \text{s.t.} \quad & c_2 + e_2 = b_2 \left[w_2 - \left(\frac{1}{\delta} \right) Dy \right] \\ & c_2 + e_2 = \omega (c_1 + e_1) \end{aligned} \quad (11)$$

where c_2 is the commodity consumption; e_2 is the education consumption; b_2 is the marginal propensity to consume; ω is the proportional distance in consumption between high-income and low-income families.

Therefore, a Lagrange function is established.

$$\varphi = c_2^\alpha e_2^{1-\alpha} - \lambda_2 \left\{ c_2 + e_2 - b_2 \left[w_2 - \left(\frac{1}{\delta} \right) Dy \right] \right\} - \lambda_3 [c_2 + e_2 - \omega (c_1 + e_1)] \quad (12)$$

The following first-order conditions are obtained (Eq. (12)).

$$\frac{\partial \varphi}{\partial c_2} = \alpha c_2^{\alpha-1} e_2^{1-\alpha} - \lambda_2 - \lambda_3 = 0 \quad (13)$$

$$\frac{\partial \rho}{\partial e_2} = (1 - \alpha)c_2^\alpha e_2^{-\alpha} - \lambda_2 - \lambda_3 = 0 \quad (14)$$

$$\frac{\partial \rho}{\partial \lambda_2} = - \left\{ c_2 + e_2 - b_2 \left[w_2 - \left(\frac{1}{\delta} \right) Dy \right] \right\} = 0 \quad (15)$$

$$\frac{\partial \rho}{\partial \lambda_3} = - [c_2 + e_2 - \omega(c_1 + e_1)] = 0 \quad (16)$$

Eqs. (13)–(16) are used to obtain

$$c_2 / e_2 = \alpha / (1 - \alpha) \quad (17)$$

$$c_2 = \alpha b_2 \left[w_2 - \left(\frac{1}{\delta} \right) Dy \right] \quad (18)$$

$$e_2 = (1 - \alpha) b_2 \left[w_2 - \left(\frac{1}{\delta} \right) Dy \right] \quad (19)$$

Consequently, the education consumption of high-income families is $e_2 = (1 - \alpha) b_2 [w_2 - (\frac{1}{\delta}) Dy]$ to maximize family utility. The education expenditures of low-income and high-income families are added up to analyze the impact of varying income gaps on social education expenditure. Social education expenditure level E is as follows since their weight is $1 - \delta$ and δ , respectively.

The relationship between social education expenditures and varying income gaps is further analyzed by deriving E with respect to ine (Eq. (20)).

$$\frac{\partial E}{\partial ine} = \left(\frac{\partial E}{\partial Dy} \right) \left(\frac{\partial Dy}{\partial ine} \right) = (1 - \alpha)(b_1 - b_2) \left(\frac{\partial Dy}{\partial ine} \right) \quad (21)$$

According to Eq. (2),

$$\frac{\partial ine}{\partial Dy} = - \frac{\left(\frac{1}{\delta} \right) w_1 + \left(\frac{1}{1 - \delta} \right) w_2}{\left[w_1 + \left(\frac{1}{1 - \delta} \right) Dy \right]^2} < 0, \text{ 则 } \frac{\partial Dy}{\partial ine} < 0 \quad (22)$$

The relationship between b_1 and b_2 is obtained from Eqs. ((2), (7), (15) and (16)).

$$\frac{b_2}{b_1} = \omega \frac{w_1 + \left(\frac{1}{1 - \delta} \right) Dy}{w_2 - \left(\frac{1}{\delta} \right) Dy} = \frac{\omega}{ine} \quad (23)$$

The relationship between the consumption propensity of high (low) income families is different for differences between ine and ω . Therefore, the relationship between income inequality and social education expenditure occurs in the following three cases.

Case 1. $\partial E / \partial ine = (1 - \alpha)(b_1 - b_2)(\partial Tr / \partial ine) > 0$ when $ine < \omega$ and $b_2 > b_1$.

There is a positive correlation between social education expenditure and income gaps when $ine < \omega$.

The gap between high-income and low-income families is not obvious when income gaps are at a low level. However, high-income families eagerly hope that this gap will be more obvious and known to others. Therefore, high-income families will increase their commodity consumption and education consumption based on the investment effects of education and the status-seeking of consumption. It makes their consumption propensity at a higher level. In contrast, low-income families do not realize the importance of the investment effects of education in improving family utility and maintaining existing social status and wealth, and they lag due to information disadvantages. Income gaps are small at this time, and the catch-up effect of consumption is insufficient. Low-income households require higher savings to cope with risks. Consequently, their consumption propensity is at a low level.

The consumption propensity of high-income families is greater than that of low-income ones at this time ($b_2 > b_1$). High-income families spend a higher proportion of education expenditures for the same unit of income. The law of diminishing marginal propensity to consume in mainstream economics is challenged. That is, its establishment needs to improve household utility, widen gaps, and seek the status of high-income households. The consumption of high-income households satisfies the law on this basis. Therefore, social education expenditures will increase when Dy causes income inequality to expand. The gap between b_2 and b_1 becomes smaller with expanding income inequality at this time. Social education expenditures increase at a decreasing rate.

Case 2. $\partial E / \partial ine = (1 - \alpha)(b_1 - b_2)(\partial Tr / \partial ine) < 0$ when $ine > \omega$ and $b_2 < b_1$.

There is a negative correlation between social education expenditures and income gaps when $ine > \omega$.

High-income families have more investment channels, and their wealth levels rise with widening income gaps. The effect of

education investment on high-income families weakens in terms of improving family utility and maintaining existing social status and wealth. Their consumption propensity declines based on satisfying the status-seeking of consumption. At this time, income gaps among low-income families are gradually widening, and the catch-up effect of consumption is increasing. They realize that the effect of education investment can improve family utility and maintain existing social status and wealth, yet alternative avenues are lacking for investment. Low-income households have an enhanced ability to cope with risks and increase consumption based on previous savings and income. Therefore, their consumption propensity increases. Law is also challenged at this time. Low-income families continue to increase their consumption propensity to catch up with high-income ones and enhance family utility.

The consumption propensity of low-income families will exceed that of high-income ones when income gaps are greater than the fixed consumption distance ($b_2 < b_1$). The proportion of high-income families spending on education consumption will be lower for the same unit of income. Therefore, social education expenditures decline when Dy causes income inequality to expand. Their consumption propensity is closer to 0 with expanding income inequality. The gap between b_1 and b_2 becomes larger, and social education expenditures decrease at an increasing rate.

Case 3. $\partial E / \partial ine = (1 - \alpha)(b_1 - b_2)(\partial Tr / \partial ine) = 0$ when $ine = \omega$ and $b_2 = b_1$.

The first-order condition for social education expenditures on income inequality is 0 when $ine = \omega$. Social education expenditures reach the highest level.

The consumption propensity of high-income and low-income families is equal when income gaps are equal to the fixed consumption distance ($b_2 = b_1$). The two types of families spend the same proportion on education expenditure for the same unit of income. Social education expenditure can only reach the highest level based on the conditions.

The theoretical hypothesis of the work can be obtained by combining the three cases.

Hypothesis: The law of diminishing marginal propensity to consume in mainstream economics may be challenged when high-income families focus on the effects of education investment and social status-seeking of consumption. The same may be true when low-income families focus on the investment effects of education and the catch-up effects of consumption. Consequently, changed income gaps have an inverted U-shaped impact on social education expenditure. The consumption propensity of high-income families is higher than that of low-income ones if income gaps are smaller than the fixed consumption distance. Widened income gaps increase social education expenditures at this time, and vice versa.

3. Data sources, variable measurement, and statistical analysis

The section provides data sources of key variables (e.g., income gaps and social education expenditure) to verify the hypothesis. The measurement of relevant variables especially key variables is described in detail, which are involved in the empirical verification based on data characteristics. Corresponding statistical analysis is carried out on the variables.

3.1. Data sources

The theoretical analysis in the work is mainly about the relationship between income gaps and social education consumption. education expenditure is mainly a micro-family behavior, and it is difficult to obtain corresponding data from existing macro levels, e.g., provincial or even prefecture-level cities. Income gaps within cities and rural areas are difficult to measure although income gaps at the provincial or prefectural level can be measured through indicators, e.g., urban-rural income gaps. Indicators are harder to acquire, e.g., the Gini coefficient which can accurately measure income inequality.

The work used district (county) level data obtained by aggregation at the micro level. Quasi-micro panel data of representative (average) households in each district and county were obtained to verify the theoretical hypothesis. This was the main marginal contribution of the work at empirical analysis.

Microdata used in the empirical verification came from CFPS. The database was officially constructed in 2010, and the research scope covered 25 provinces/municipalities/autonomous regions except Inner Mongolia, Hainan, Tibet, Qinghai, Ningxia, and Xinjiang. The target sample size is 16,000 households, and the survey respondents include all household members in the sample households. CFPS conducted test surveys for initial and follow-up visits in Beijing, Shanghai, and Guangdong in 2008 and 2009, respectively, and formally launched the interviews in 2010. All the baseline household members defined by the 2010 baseline survey and their future blood/adopted children will be the genetic members of CFPS and become the permanent tracking objects. There are four main questionnaire types in CFPS, namely, Community Questionnaire, Family Questionnaire, Adult Questionnaire, and Children Questionnaire, and based on these, long questionnaires, short questionnaires, and questionnaires on behalf of the family members of different natures are constantly being developed. On the basis of this, long questionnaires, short questionnaires, proxy questionnaires, e-interview questionnaires and other questionnaire types have been developed for family members of different nature. It reflected the changes in China's society, economy, population, education, and health and provided micro-data support for academic research and public policy analysis. The collection and release of data were completed in cooperation with the Institute of Social Science Survey, Peking University, and the Survey Research Center of the University of Michigan.

The empirical analysis mainly used survey data from 2012 to 2018 to construct quasi-micro panel data. Data content included each issue's family questionnaire database, adult questionnaire database, and children's questionnaire database. We matched the samples and eliminated districts (counties) with missing values or a valid number of households less than 50 to make sample data more scientific. The work finally obtained valid sample data for a total of 470 districts (counties) in 4 periods.

3.2. Variable measurement and descriptive statistics

Social education expenditure was used as the explained variable. The work utilized the mean value of education expenditure in households within the district (county) as a proxy for the social education consumption level in the region. "How much was your family's education expenditure in the past 12 months?" was a question in the CFPS household questionnaire. The respondents' answers

Table 1
Variable definitions and descriptive statistics.

Variable	Definition	Observed number	Average	Standard deviation	Minimum	Medium	Maximum
Overall educational spending	The logarithm of the average household education expenditure in a district (county). The average household education expenditure in a district (county) is the sum of the education expenditures of families participating in the survey in the district (county) divided by the number of families participating in the survey. It is used as the representative or average family education expenditure.	470	7.030	2.110	0.000	8.025	9.560
Income gap	Gini coefficient of district (county) income	470	0.463	0.080	0.273	0.456	0.718
Government subsidies	The logarithm of the mean government subsidy for representative (average) households in a district (county). The mean is calculated in the same way as the overall education expenditure.	470	6.511	0.778	2.034	6.547	9.014
Social donations	The logarithm of the mean social contribution of representative (average) households in a district (county). The mean is calculated in the same way as the overall education expenditure.	470	4.671	1.703	0.000	4.369	9.085
Property ownership	The average property ownership of representative (average) households in the district (county). The average is calculated in the same way as the overall education expenditure. (Unit: set)	470	0.186	0.111	0.011	0.168	0.740
Cash and deposits	The logarithm of the mean cash and savings value of representative (average) households in a district (county). The mean is calculated in the same way as the overall education expenditure.	470	10.086	0.809	7.751	10.002	13.075
Financial products	The logarithm of the mean value of the total value of financial products of representative (average) households in a district (county). The mean is calculated in the same way as the overall education expenditure.	470	5.977	4.196	0.000	7.422	12.739
Family size	The mean value of the representative (average) family size of the district (county), the mean is calculated in the same way as the overall education expenditure	470	3.822	0.740	2.183	3.745	6.800
Self-employment	District (county) representative (average) household self-employment rate (engagement = 1; otherwise, 0)	470	0.098	0.055	0.000	0.089	0.315
housing rental	District (county) representative (average) household housing rental rate, calculated in the same way as an overall education expenditure (rent = 1; otherwise, 0)	470	0.038	0.055	0.000	0.017	0.328
Age	The average age of household heads in a representative (average) district (county). The average is calculated in the same way as the overall education expenditure (unit: year)	470	49.994	4.320	38.590	50.207	60.765
Gender	The gender rate of representative (average) household heads in the district (county), calculated in the same way as an overall education expenditure (male = 1; female = 0)	470	0.516	0.108	0.224	0.508	0.889
Education level	The average number of education years received by the head of a representative (average) household in a district (county). The average is calculated in the same way as the overall education expenditure (unit: year)	470	7.002	1.697	1.385	7.012	11.412
Employment	The work rate of the representative (average) household head in the district (county), calculated in the same way as the overall education expenditure (Yes = 1; otherwise, 0)	470	0.734	0.136	0.196	0.752	0.987
Health	The average health level of the representative (average) household heads in the district (county) is calculated in the same way as the overall education expenditure (very healthy = 5; quite healthy = 4; rather healthy = 3; fairly healthy = 2; unhealthy = 1)	470	1.814	0.259	0.952	1.817	2.600

referred to the family's education expenditure in the past year. As survey data, CFPS could not survey all households in a district or county. Therefore, the social education expenditure was not obtained by adding up the education expenditures of all households in a district or county.

CFPS considered households with different characteristics within a district (county) as China's authoritative micro survey data. The work excluded districts or counties with fewer than 50 effective households to ensure the integrity of data organization and sample reconstruction. Typical low-income and high-income households were included. The education expenditures of surveyed households were averaged (the sum of the education expenditures of surveyed households in the region was divided by the participating households) when the social education expenditure in the region was not obtained by adding up the education expenditures of all households in the region. The optimal choice under the given constraints was to obtain the education expenditure of a representative (average) family based on the number of families surveyed by averaging that.

Using the average could avoid scaling deviations caused by the different number of households sampled in each district (county) during the sampling survey process. For example, the high education expenditure in a specific district (county) was attributed to the substantial participation of families from that district (county) in the sample survey instead of the income disparity. More importantly, the social education expenditure was obtained by the weighted average of the education expenditures of low-income and high-income families in the theoretical analysis. Therefore, the work utilized the mean of education expenditure in households within the district (county) as a proxy for the social education consumption level in the region. It was highly consistent with the theoretical hypothesis.

The income gap was the core explanatory variable in the work. The most commonly used indicator was the Gini coefficient when the income gap is measured. Because the Gini coefficient gives a quantitative boundary reflecting the degree of difference between the rich and the poor among residents, it can reflect and monitor the gap between the rich and the poor among residents in a more objective and intuitive manner, and forecast, warn and prevent the polarisation of the rich and the poor among residents. And the measured Gini coefficient can be compared with the nationally published Gini coefficient to prevent the influence of outliers. The work used this indicator to reflect income inequality. Samples did not accurately reflect regional income distribution due to the small sample size in each community (village) of CFPS. The sample size at the provincial level was sufficient, but the coverage was macro. Some uncontrollable factors could be included in calculating regression. Therefore, the Gini coefficient was calculated in the district (county) (Liu & Wang, 2017; Zhou et al., 2014).

Existing research points out that income cannot accurately measure a family's ability to use economic resources in developing countries for some hidden income and benefits omitted from the statistics. Therefore, the expenditure can more accurately reflect the number of economic resources controlled by a family compared with income (Strauss & Thomas, 2008). Consequently, the work used percentile expenditure ratios P90/P10 and P90/P50 instead of the Gini coefficient to measure the district (county) income gap in the robustness analysis. P90/P10 referred to the ratio of household expenditures at the 90th percentile to the 10th percentile; P90/P50 referred to the ratio of household expenditures at the 90th percentile to the 50th percentile. Both measured the income gap between high-income households and varying degrees of low-income households.

Other variables affecting regional education expenditures are controlled to obtain varying income gaps in social education expenditures and solve regression bias caused by omitted variables. The work controlled a series of characteristic variables of regionally representative (average) households and their head according to the research of Chen et al. in 2014 and Wu et al. in 2018. Regionally representative (average) family characteristic variables mainly include government subsidies, social donations, property ownership, cash and deposits, financial products, family size, self-employment, and housing rental. Characteristic variables of regionally representative (average) household heads mainly include age, gender, education level, work, and health status. The control variables are calculated consistently with the explained variables, taking into account the education expenditure of regionally representative families. The sum of the corresponding indicators in the district (county) is divided by the number of households in the corresponding district (county). The health of the household head is calculated from the household head's responses to the question "How do you think your health is?" (Very healthy = 5, Quite healthy = 4, Rather healthy = 3, Fairly healthy = 2, and Unhealthy = 1). CFPS data do not provide household head information. Financial respondent information is used as a substitute for household head information. Table 1 shows the definitions and measurement indicators of relevant variables.

A quasi-micropanel data sample of representative (average) households in a district (county) is obtained based on the variable selection and indicator construction. Cross-sectional individuals in each year are the districts (counties) involved in the survey. Districts (counties) surveyed each year are unlikely to change significantly although households surveyed by CFPS are randomly selected. Therefore, panel data with stable cross-sectional individuals are obtained.

Micro-data of representative (average) households in the district (county) are calculated in the indicator construction. Used indicators are household microscopic ones although cross-sectional individuals are macroscopic districts (counties). Household micro-panel data are obtained at the district (county) level, which is called quasi-micro panel data in the work. Notably, there are slight changes in the (districts) and counties surveyed each year; thus, obtained quasi-micro panel data is further processed.

Districts (counties) appearing in each year are taken as separate samples to form strong panel data (with a sample size of 312). Cross-sectional individuals in data are the same for each year, except for those that are valid but do not appear annually. All effective districts (counties) are included in the sample to form weak panel data (with a sample size of 470) to maximize effective information. Cross-sectional individuals in each year are not necessarily all the same (most are the same). Therefore, all samples are fully utilized. The sample size of the weak panel data is 470, which exceeds that of strong panel data by 158 and represents a 50% increase. Therefore, more robust conclusions are obtained.

A statistical analysis of relevant variables is conducted based on quasi-micropanel data (Table 1). Note that descriptive statistics are based on weak panel data with 470 samples. The maximum logarithm of education expenditure of representative (average) households in districts and counties is 9.560, with a minimum of 0 and a standard deviation of 2.11. There is a large gap among different

representative (average) households concerning education expenditures. Similar characteristics exist in logarithms of government subsidies, social donations, cash and deposits, and financial products for representative (average) households in districts and counties.

The mean (7.03) of the logarithm of representative (average) household education expenditures in districts and counties is less than the median (8.025), indicating that the education expenditures of most residents in China are generally low and far from the median level. The mean of the district (county) Gini coefficient is 0.463 while the median is 0.456, indicating that a huge income gap in most districts (counties). The mean of house ownership by representative (average) households in districts and counties is 0.186 suite with a maximum of 0.74 suite. The representative (average) household property ownership in most districts (counties) does not reach 1 suite.

The mean of representative (average) family size in districts and counties is 3.822, while the median is 3.745, indicating that the population of a representative (average) family in most districts (counties) is three to four. The mean of self-employment among representative (average) households in districts and counties is 0.098, indicating that most representative (average) households in districts (counties) are not engaged in self-employment. The mean of housing rental for representative (average) families in districts and counties is 0.038, indicating that most representative (average) families in districts (counties) do not have additional houses for rent. It is consistent with statistical results of representative (average) household property ownership in districts and counties.

The statistical findings from a representative sample of household heads in the region indicate that the majority of district-level household heads are approximately 50 years old, employed, and generally exhibit a level of health slightly below average (with a mean score of 1.814 on a scale ranging from unhealthy to fairly healthy). Furthermore, these individuals have received limited education, with an average duration of schooling being 7 years, indicating completion of primary education. An interesting phenomenon is that the gender average of regionally representative (average) household heads is 0.516, showing that the proportion of male and female household heads in China is roughly the same. The hostess of a family in modern society has more control over financial management compared to the traditional "man is the head of a family" by combining the household head identity of the "financial respondent".

Table 2
Baseline regression.

Variable	(1) Strong panel	(2) Strong panel	(3) Weak panel	(4) Weak panel
	EC	EC	EC	EC
GINI	−1.680** (0.762)	13.050*** (2.487)	−0.251 (0.228)	3.678** (1.481)
GINI ²		−15.579*** (3.121)		−4.035*** (1.528)
Government subsidies	−0.012 (0.041)	−0.013 (0.042)	0.074*** (0.009)	0.070*** (0.010)
Social donations	0.033** (0.012)	0.034*** (0.011)	0.027*** (0.005)	0.028*** (0.006)
Property ownership	1.028** (0.417)	1.048** (0.408)	0.417*** (0.095)	0.371*** (0.088)
Cash and deposits	0.119 (0.079)	0.147** (0.071)	0.229** (0.111)	0.236** (0.110)
Financial products	−0.022* (0.013)	−0.020* (0.011)	−0.040** (0.020)	−0.040** (0.019)
Family size	0.005 (0.112)	0.008 (0.109)	0.171*** (0.025)	0.172*** (0.025)
Self-employment	1.013 (0.867)	0.926 (0.858)	0.360 (0.481)	0.309 (0.479)
Housing rental	−1.698 (1.471)	−1.950 (1.452)	−2.436* (1.371)	−2.398* (1.367)
Age	−0.008 (0.009)	−0.008 (0.009)	0.009** (0.004)	0.008* (0.005)
Gender	0.941 (0.615)	0.887 (0.619)	0.920* (0.473)	0.922* (0.471)
Education level	0.149*** (0.028)	0.138*** (0.030)	0.226*** (0.056)	0.227*** (0.056)
Work	−0.671** (0.318)	−0.890** (0.362)	−1.219** (0.535)	−1.291** (0.538)
Health status	−0.130*** (0.024)	−0.140*** (0.022)	−0.354* (0.194)	−0.356* (0.195)
Constant term	7.192*** (0.566)	3.845*** (0.252)	3.920*** (0.839)	3.078*** (0.903)
Sample size	312	312	470	470
Goodness of fit	0.881	0.883	0.858	0.858
Group number	78	78	143	143
F-statistics	9.103	20.96	19.48	2.877
Hausman-test	Statistics : 458.588		P = 0.00 < 0.05	
F-test	Statistics : 2.920		P = 0.00 < 0.05	

Note: ***, **, and * indicate significance at the 1, 5, and 10%, respectively. Robust standard errors are in parentheses.

4. Model and analysis of empirical results

An econometric model is constructed in the section based on quasi-micropanel data. Endogenous analysis and robustness testing are performed based on the baseline regression. The theoretical hypothesis is empirically verified. Besides, the work preliminarily determines the inflection point of the inverted U-shaped curve and its relationship with the critical value (the fixed consumption distance between high-income and low-income households).

4.1. Model construction

There is an inverted U-shaped relationship between varying income gaps and social education expenditures according to the theoretical hypothesis. Therefore, the panel-data econometric model is defined by

$$EC_{it} = \beta_0 + \beta_1 GINI_{it} + \beta_2 GINI_{it}^2 + \beta_3 X_{it} + \mu_i + \varepsilon_{it}$$

where EC_{it} is the education expenditure level of representative (average) households of district (county) i in the t th period; $GINI_{it}$ is the Gini coefficient of district (county) i in the t th period; X_{it} is the control variable at the household head and the representative (average) of district (county) levels; μ_i is the unobservable individual fixed effect; ε_{it} is the random disturbance term.

4.2. Results of baseline regression

P of relevant statistics of the baseline-regression model is less than 0.05 from Hausman and BPLM tests; thus, the model of individual fixed effect is selected to estimate quasi-micro panel data. Table 2 shows baseline regression results. Models (1) and (2) represent stepwise regression based on strong panel data, while models (3) and (4) are based on weak panel data. There is a nonlinear relationship between income inequality and regional education expenditure levels; thus, the theoretical hypothesis is reasonable. Model (4) contains the most samples, more comprehensive control variables, and more significant coefficient estimation of related variables. Therefore, the results of model (4) are used to explain.

According to model (4), the estimated coefficient of the quadratic term of the Gini coefficient is -4.035 , which is significant at the 1% significance level. There is a significant inverted U-shaped relationship between the income gap and overall education expenditure. Expanding income gaps promote overall education expenditure before the income gap reaches its peak (Section 5). It declines the overall education expenditure after the income gap reaches its peak. This preliminarily verifies the theoretical hypothesis according to the coefficient estimation of linear and quadratic terms of the Gini coefficient in model (4).

The peak of inverted U-shaped curves is 0.455. It is smaller than the average Gini coefficient of each district (county) (0.463) and extremely close to the median (0.456) (Table 1). More than half of districts (counties) have serious income inequality in the current sample period, which has an inhibitory effect on overall education expenditure in the region. Therefore, the income gap should be narrowed to the peak (0.455) to increase the overall education expenditure level. The conclusion indicates that China's current income gaps are overlarge for the whole society.

Common prosperity to narrow income gaps does not have a significant inhibitory effect on education expenditure and human capital accumulation. Narrowing income gaps to a critical value maximizes social education expenditures, which promotes a high-quality economy. Common prosperity cannot be achieved overnight. It must be carried out step by step and in an orderly manner based on the trade-off between fairness and efficiency. Currently, narrowing income gaps is consistent with fairness and efficiency compared with overlarge income gaps.

Estimation results of model (4) are used to analyze the coefficient estimation of control variables. The coefficient estimations of financial products and housing rental are significantly negative at 10% within regional representative (average) household characteristic variables. It shows that the more the regional representative (average) household financial product and housing rental, the lower the overall education expenditure level. The more financial products and housing rentals of regional representative (average) households, the lower their dependence on labor income and their interest in education investment to improve the human capital for high future incomes. Therefore, the overall education expenditure is lower.

Government subsidies, social donations, property ownership, cash and deposits, and family size of representative (average) families significantly contribute to the increase in education expenditures. The effect of self-employment is positive insignificantly. The coefficient estimations of the household head's work and health are negative at the 10% significance level in characteristics variables of regional representative (average) households. The higher the working rates and health levels of the head of a regionally representative (average) household, the lower the overall education consumption level. It may be because their high working rates and health levels enable them to have stable incomes, higher living standards, and lower living burdens. Their expectation to obtain higher income and improve family status through education weakens; thus, the overall education expenditure level becomes lower. The age, gender, and educational level of the household head in other variables significantly contribute to the household's education expenditure.

There are two significant variables in control variables with positive effects—government subsidies and the educational level of the household head. Increasing government subsidies and the education level of household head can increase education expenditures. Therefore, we should focus on increasing government subsidies for families with higher education levels to maximize education expenditures.

4.3. Endogenous analysis

In general, the previous period of the endogenous variable is not correlated with the error term in the current period, so one can consider using a lagged period of the endogenous variable in place of the endogenous variable in the current period. The work performs first-order lag processing on explanatory variables to alleviate the endogeneity from bidirectional causation and explore the possible time lag in the impact of independent variables on dependent ones. Table 3 shows the results after stepwise regression. An inverted U-shaped correlation exists between income gaps and regional education consumption at the 1% significance level (model (6)).

The education level of household heads is significantly positive, while government subsidies are insignificantly positive in relevant control variables. The work and health of household head are consistent with baseline-regression results and have a significantly negative impact on education expenditures. The coefficient estimations for financial products and housing rentals are significantly negative, indicating the robustness of the regression results of the baseline model to some extent.

4.4. Robustness test

1) Using percentile expenditure ratio to measure income gaps. Income cannot accurately measure a family's ability to use economic resources in developing countries for some hidden income and benefits omitted from the statistics. Therefore, the expenditure can more accurately reflect the number of economic resources controlled by a family compared with income. District (county) percentile expenditure ratios P90/P10 and P90/P50 are used for robustness tests to further verify the impact of income inequality on the overall education consumption level. Coefficients of quadratic terms of P90/P10 and P90/P50 are significantly negative (Table 4), indicating a significant inverted U-shaped relationship between varying income gaps and overall regional education expenditure.

The effects of government subsidies and the education level of household head are significantly positive, while the effects of financial products and the job and health of household head are significantly negative. The impact of housing rentals is negative insignificantly, indicating robust estimation results of the baseline regression model. The work uses indicators at income and

Table 3
Endogenous analysis.

Variable	(5) Lag for a period	(6) Lag for a period
	EC	EC
GINI	−0.872 (1.270)	5.704*** (0.846)
GINI ²	0.119 (1.539)	−5.927*** (0.883)
Government subsidies		0.040 (0.040)
Social donations		−0.003 (0.009)
Property ownership		0.004 (0.016)
Cash and deposits		0.020 (0.029)
Financial products		−0.004*** (0.001)
Family size		0.205*** (0.011)
Self-employment		−0.248*** (0.064)
Housing rental		−1.266*** (0.241)
Age		−0.003 (0.005)
Gender		−0.024 (0.123)
Education level		0.116*** (0.002)
Work		−0.434*** (0.074)
Health status		−0.040** (0.019)
Constant term	8.460*** (0.249)	5.271*** (0.774)
Sample size	299	299
Goodness of fit	0.114	0.359
Group number	121	121
F-statistics	13.13	124.4

Note: ***, **, and * indicate significance at the 1, 5, and 10%, respectively. Robust standard errors are in parentheses.

Table 4
Robustness test using percentile expenditure ratios to measure income gaps.

Variable	(7)	(8)
	EC	EC
P90/P10	0.069** (0.027)	
(P90/P10) ²	−0.003** (0.001)	
P90/P50		0.659*** (0.151)
(P90/P50) ²		−0.076** (0.029)
Government subsidies	0.049*** (0.009)	0.037*** (0.014)
Social donations	0.029*** (0.001)	0.027*** (0.002)
Property ownership	0.468*** (0.156)	0.480*** (0.152)
Cash and deposits	0.175* (0.100)	0.146 (0.093)
Financial products	−0.038** (0.019)	−0.038** (0.017)
Family size	0.147*** (0.035)	0.155*** (0.037)
Self-employment	0.652 (0.678)	0.462 (0.655)
Housing rental	−1.870 (1.307)	−1.792 (1.280)
Age	0.014* (0.008)	0.011 (0.008)
Gender	0.957** (0.471)	0.761* (0.405)
Education level	0.218*** (0.051)	0.224*** (0.049)
Work	−1.048*** (0.392)	−1.103*** (0.371)
Health status	−0.242** (0.097)	−0.232** (0.090)
Constant term	3.657** (1.468)	3.373*** (1.210)
Sample size	449	449
Goodness of fit	0.869	0.871
Group number	140	140
F-statistics	19.12	9.255

Note: ***, **, and * indicate significance at the 1, 5, and 10%, respectively. Robust standard errors are in parentheses.

expenditure levels to measure income gaps. Regression results corroborate each other, which verifies the scientific nature of the theoretical hypothesis.

2) Bilateral winsorization. The data chosen for this paper is questionnaire data, which may have extreme or problematic data. Relevant continuous variables are usually winsorized in regression analysis to prevent outliers from affecting results. The work also performs stepwise regression on explanatory variables, explained variables, and control variables at 1 and 99% percentiles after winsorizing them. Table 5 shows stepwise regression results, and the results of model (10) are consistent with the previous statement. There is a significant inverted U-shaped relationship between income gaps and education expenditure. The impact of government subsidies and the education level of the household head on education expenditure is significantly positive, while that of financial products, housing rentals, and household head's health and work is significantly negative. Results are completely consistent with baseline regression results, further verifying the robustness of the baseline regression model and the scientific nature of the theoretical hypothesis.

4.5. Summary and comparison of inflection points and their relationship with critical values

Models (2), (4), (6), (7), (8), and (10) are used for baseline regression, endogenous analysis, and robustness tests on the inverted U-shaped relationship between varying income gaps and household education expenditures by introducing control variables. Results verify the theoretical hypothesis. However, we are concerned about the non-linear relationship and the economic meaning represented by the inflection point and its relationship with the critical value in the theoretical hypothesis (the fixed consumption distance between high-income and low-income households).

Table 5
Robustness test by bilateral winsorization.

Variable	(9)	(10)
	EC_w	EC_w
GINI_w	−0.322 (0.218)	3.517** (1.508)
GINI ² _w		−3.959** (1.550)
Government subsidies_w	0.065*** (0.014)	0.061*** (0.015)
Social donations_w	0.028*** (0.006)	0.029*** (0.007)
Property ownership_w	0.413*** (0.100)	0.372*** (0.094)
Cash and deposits_w	0.233* (0.119)	0.240** (0.119)
Financial products_w	−0.041** (0.020)	−0.041** (0.020)
Family size_w	0.156*** (0.031)	0.157*** (0.031)
Self-employment_w	0.422 (0.493)	0.375 (0.494)
Housing rental_w	−2.566* (1.466)	−2.527* (1.457)
Age_w	0.008** (0.003)	0.007** (0.004)
Gender_w	0.967* (0.500)	0.969* (0.498)
Education level_w	0.221*** (0.056)	0.222*** (0.056)
Work_w	−1.225** (0.570)	−1.292** (0.571)
Health status_w	−0.376* (0.213)	−0.377* (0.213)
Constant term	4.144*** (0.723)	3.306*** (0.770)
Sample size	470	470
Goodness of fit	0.857	0.858
Group number	143	143
F-statistics	12.86	3.171

Note: ***, **, and * indicate significance at the 1, 5, and 10%, respectively. Robust standard errors are in parentheses.

The inflection points of model (4) have been analyzed, but those of the inverted U-shaped curves in all relevant models should be summarized and compared. Model (6) of the endogenous analysis introduces lag terms of explanatory variables, which cannot represent the impact of the current income gap on education expenditure in economics. Therefore, the section mainly summarizes and compares inflection points of inverted U-shaped curves from models (2), (4), (7), (8), and (10) and their relationships with critical values in the theoretical hypothesis. Table 6 lists relevant results.

The inflection point (peak) of the inverted U-shaped curve obtained by measuring the income gap using the Gini coefficient is not a critical value, but it has a one-to-one correspondence with that. The current income gap measured by most districts (counties) using different average household income ratios exceeds the critical value if that exceeds the peak of the inverted U-shaped curve, and vice versa. The following is a detailed discussion based on the analysis and results (Table 6).

The peak of the inverted U-shaped curve of model (4) is the largest among models (2), (4), and (10). The peak of models (2) and (10) are 0.419 and 0.445, respectively, which are smaller than the median (0.456) and mean (0.463) of the Gini coefficient. The income gap in most districts (counties) in China is overlarge during the sample period compared to maximizing education expenditure.

The fixed consumption distance between high-income households at the 90th percentile and low-income ones at the 10th percentile is 11.5 according to model (7). That between high-income households at the 90th percentile and low-income ones at the 50th percentile is 4.3. Therefore, there are obvious differences in consumption levels among different income classes in China, which are reflected in the consumption of general commodities and education.

The fixed consumption distance between high-income households at the 90th percentile and low-income ones at the 50th percentile is smaller ($11.5/4.3 = 2.7$) from $P50/P10 = (P90/P10)/(P90/P50)$. The fixed consumption distance between the 90th and 50th percentiles (4.3) is 1.6 times that between the 50th and 10th percentiles (2.7). The income gap is overlarge in China. The former and the latter should be consistent if the income gap is reasonable.

In summary, there is a one-to-one correspondence between peaks and critical values of inverted U-shaped curves from models (2), (4), (7), (8), and (10) and income gaps measured by different methods (Gini coefficient, different households' average income and consumption ratios, etc.). Currently, the income gap in most districts (counties) in China exceeds the critical value. Therefore, common

Table 6

Inflection points of inverted U-shaped curves in relevant models and their relationship with the critical value.

Variable	Model (2)	Model (4)	Model (7)	Model (8)	Model (10)
Gini coefficient quadratic term	−15.579	−4.035			−3.959
Gini coefficient linear term	13.050	3.678			3.517
Percentile expenditure ratio (P90/P10) quadratic term			−0.003		
Percentile expenditure ratio (P90/P10) one-time term			0.069		
Percentile expenditure ratio (P90/P50) quadratic term				−0.076	
Percentile expenditure ratio (P90/P50) one-time term				0.659	
Inflection points (Critical value or not)	0.419 (No)	0.455 (No)	11.5 (Yes)	4.3 (Yes)	0.445 (No)
Gini coefficient mean	0.463	0.463			0.463
Gini coefficient median	0.456	0.456			0.456

From the authors.

Models (2), (4), and (10) use the Gini coefficient to measure income gaps. Inflection points of the inverted U-shaped curve obtained by the model can thus be compared with the mean and median of the Gini coefficient. It determines whether income gaps among various districts (counties) in China during the sample period are overlarge (on the right side of the inverted U-shaped curve) or small (on the left side of the inverted U-shaped curve). The flaw is that the Gini coefficient is calculated differently from the critical value in the theoretical hypothesis (the fixed consumption distance between high-income and low-income households); thus, the inflection point obtained is not the critical value in the theoretical hypothesis.

Models (7) and (8) use the consumption ratio of households at different percentiles to measure the income gap. The higher the percentile, the higher the income, and therefore the higher the consumption. The inflection point of the inverted U-shaped curve is the critical value in the theoretical hypothesis. Those in the 50th and 10th percentiles belong to low-income families compared with high-income families in the 90th percentile according to the hypothesis. However, the indicator cannot determine whether China's current income gaps are overlarge or small. This is because income gaps and critical value need to be compared according to the theoretical analysis. Indicators to measure income gaps in the empirical analysis are the Gini coefficient (baseline regression) and the consumption ratio of different households (robustness test). The ratio of the average income of high-income families to low-income ones is not used in the theoretical analysis (see the hypothesis).

Models (2), (4), and (10) need analyzing together with models (7) and (8). It thus determines whether China's current income gaps are overlarge or small as well as corresponding critical values. Each Gini coefficient corresponds to each average income ratio of different families for the income gap in the established framework of analysis and data sample. Each average income ratio of different households corresponds to each ratio of consumption expenditures of different households, and vice versa.

prosperity to narrow income gaps has become an inevitable choice for increasing social education expenditures. It does not have a significant inhibitory effect on education spending and human capital accumulation. Narrowing income gaps to a critical value maximizes social education expenditure to promote an overall high-quality economy.

5. Conclusions and policy implications

China's income gaps will narrow in the long run considering common prosperity. What effects do decreased income gaps have on promoting a high-quality economy? The work initially answered the question by focusing on education expenditure. It was a form of consumption that has great significance to the future development of residents and overall social and technological innovation.

The nonlinear relationship between the two was empirically verified based on CFPS quasi-micro panel data. The law of diminishing marginal propensity to consume in mainstream economics could be challenged when high-income families focused on the investment effects of education and social-status seeking of consumption. The same could be true when low-income countries focused on the investment effects of education and the catch-up effects of consumption. Consequently, changed income gaps had an inverted U-shaped impact on social education expenditure. The consumption propensity of high-income families was higher than that of low-income ones if income gaps were smaller than the fixed consumption distance. Widened income gaps increased social education expenditure at this time, and vice versa.

The empirical analysis further found that income gaps in most districts (counties) in China exceeded the critical value during the sample period based on the verification of an inverted U-shaped relationship. China's current income gaps were overlarge. Therefore, narrowed income gaps by common prosperity had no significant inhibitory effect on education expenditure and human capital accumulation. The critical value would maximize overall education expenditure to promote a high-quality economy in China. Government subsidies and the education level of household heads significantly increased residents' education expenditures. Therefore, we should focus on increasing government subsidies for families with higher education levels.

In summary, current excessive income gaps are inconsistent with fairness and efficiency. Common prosperity to narrow income gaps (before the critical value) is consistent with fairness and efficiency. It is undeniable that there will be a contradiction between socialist equity (narrowing income gaps) and the efficiency of the market economy (maximizing social education expenditures) with gradual common prosperity and narrowing income gaps. Therefore, the traditional free market economic system is incompatible with China's socialist political system of common prosperity.

The decisive role of the market in resource allocation and the government should be given full play, and effective markets should be organically combined with a responsible government. Only then can common prosperity be achieved with the economy promoted. Common prosperity cannot be achieved overnight. It must find a balance between fairness and efficiency. Distribution should be rational and scientific when the overall economy is developed to the maximum extent; thus, the majority of residents, especially low-income families, can share the fruits of economic development.

CPC faces more non-Pareto-improvement policy choices with narrowing income gaps. Conflicts of interest among different classes and groups come to the fore. It is necessary to further strengthen the core leadership of CPC and the historical status of its people. Consequently, common prosperity can be achieved in China's socialist modernization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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